Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **marks**.

Solutions

1. Swap Two Numbers

Question:
Write a Python program to read two numbers and swap their values without using a third variable. Display the values before and after swapping.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("Before swapping:")
print("a =", a)
print("b =", b)

a = a + b
b = a - b
a = a - b

print("After swapping:")
print("a =", a)
print("b =", b)
```

2. Voting Eligibility

Question:
Write a Python program to read a person's age and determine whether the person is eligible to vote. If the age is negative, display an appropriate message.

```
age = int(input("Enter age: "))

if age < 0:
    print("Invalid age")
elif age >= 18:
    print("Eligible to vote")
else:
    print("Not eligible to vote")
```

3. Extract Digits

Question:
Write a Python program to read a three-digit number and display its hundreds, tens, and units digits separately.

```
n = int(input("Enter a three-digit number: "))

hundreds = n // 100
tens = (n // 10) % 10
units = n % 10

print("Hundreds digit:", hundreds)
print("Tens digit:", tens)
print("Units digit:", units)


price = float(input("Enter the price: "))


if price >= 1000:
```

```
    discount = price * 10 / 100
else:
    discount = price * 5 / 100


final_price = price - discount


print("Discount amount:", discount)
print("Final price:", final_price)
```

5. Electricity Bill

Question:
Write a Python program to read the number of electricity units consumed and calculate the bill using the following tariff:

- First 100 units: ₹2 per unit
- Next 100 units: ₹3 per unit
- Above 200 units: ₹5 per unit

```
units = int(input("Enter the number of units: "))

if units <= 100:
    bill = units * 2
elif units <= 200:
    bill = (100 * 2) + ((units - 100) * 3)
else:
    bill = (100 * 2) + (100 * 3) + ((units - 200) * 5)
```

```
print("Electricity bill: ₹", bill)
```

6. Character Classification

Question:
Write a Python program to read a character and determine whether it is an uppercase letter, lowercase letter, digit, or special character.

```
ch = input("Enter a character: ")

if ch >= 'A' and ch <= 'Z':
    print("Uppercase letter")
elif ch >= 'a' and ch <= 'z':
    print("Lowercase letter")
elif ch >= '0' and ch <= '9':
    print("Digit")
else:
    print("Special character")
```

7. Divisibility Check

Question:
Write a Python program to read an integer and determine whether it is divisible by 3, divisible by 5, divisible by both 3 and 5, or divisible by neither.

```
n = int(input("Enter an integer: "))

if n % 3 == 0 and n % 5 == 0:
    print("Divisible by both 3 and 5")
elif n % 3 == 0:
```

```
    print("Divisible by 3")
elif n % 5 == 0:
    print("Divisible by 5")
else:
    print("Divisible by neither 3 nor 5")
```

8. Sum of Digits

Question:
Write a Python program to read an integer and calculate the sum of its digits using a while loop.

```
n = int(input("Enter an integer: "))

if n < 0:
    n = -n

sum_digits = 0

while n > 0:
    digit = n % 10
    sum_digits = sum_digits + digit
    n = n // 10

print("Sum of digits:", sum_digits)
```

9. Numbers with a Condition

Question:
Write a Python program to read two integers representing a starting and ending value and display all numbers in that range that are divisible by 3 but not by 5.

```
start = int(input("Enter starting value: "))
end = int(input("Enter ending value: "))

for n in range(start, end + 1):
    if n % 3 == 0 and n % 5 != 0:
        print(n)
```

10. Value, Type and ID

Question:
Write a Python program to read an integer from the user and display its value, data type using type(), and memory identity using id(). Then add 10 to the value and display the resulting value, data type, and ID.

```
n = int(input("Enter an integer: "))

print("Original value:", n)
print("Original type:", type(n))
print("Original ID:", id(n))

n = n + 10

print("New value:", n)
print("New type:", type(n))
print("New ID:", id(n))
```

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